WORKED SOLUTION

STEP I 2010, Question 8

Pure • Infinite Descent

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q8

The key idea

Both parts use the same trick, known as *infinite descent*. First we look at the equation modulo a prime (3 in part (i), 5 in part (ii)) and find that the prime is forced to divide every unknown. Dividing it out then gives a smaller solution of *exactly the same equation*, and a non-zero integer can't be divided by 3 (or 5) for ever. The only real work is spotting which residues are possible, and in part (ii) the final digit of a fourth power does that for us.

Solution

Part (i)

Rearranging,

$$a^3 = 9c^3 - 3b^3 = 3(3c^3 - b^3),$$

so 3 divides a^3. Since 3 is prime, if it divides a product $a \cdot a \cdot a$ then it divides one of the factors, so $3 \mid a$.

Write $a = 3a_1$ with a_1 an integer. Substituting and dividing by 3:

$$27a_1^3 + 3b^3 = 9c^3 \quad \Longrightarrow \quad 9a_1^3 + b^3 = 3c^3,$$

so $b^3 = 3(c^3 - 3a_1^3)$ is divisible by 3, and as before $3 \mid b$. Write $b = 3b_1$; then

$$9a_1^3 + 27b_1^3 = 3c^3 \quad \Longrightarrow \quad c^3 = 3(a_1^3 + 3b_1^3),$$

so $3 \mid c$ too. This shows that a, b and c are all divisible by 3.

Now write $c = 3c_1$ as well. The last equation becomes $27c_1^3 = 3a_1^3 + 9b_1^3$, that is,

$$a_1^3 + 3b_1^3 = 9c_1^3.$$

Ask yourself We have shown 3 divides everything. Why does that force everything to be zero?

Because $(a_1, b_1, c_1) = (a/3,\ b/3,\ c/3)$ is again an integer solution of the *original* equation, so everything we have just proved applies to it: a_1, b_1, c_1 are divisible by 3, i.e. a, b, c are divisible by 9. Repeating the argument k times shows that 3^k divides each of a, b and c, for every $k \geq 1$. But a non-zero integer n is not divisible by 3^k once $3^k > |n|$. Hence $a = b = c = 0$, and this clearly is a solution, so it is the only one.

(Equivalently: if there were a solution other than $(0,0,0)$, choose one with $|a| + |b| + |c|$ as small as possible but positive; then $(a/3, b/3, c/3)$ is a solution with a smaller positive value, a contradiction.)

Part (ii)

Since $p^4 = |p|^4$, the final digit of p^4 depends only on the final digit of $|p|$:

last digit of n	0	1	2	3	4	5	6	7	8	9
last digit of n^4	0	1	6	1	6	5	6	1	6	1

(For example $2^4 = 16$, $3^4 = 81$, $7^4 = 2401$.) So a fourth power ends in 0, 1, 5 or 6, and it ends in 0 or 5 exactly when $5 \mid n$.

Now look at the final digit of each term.

- $5r^4$ is a multiple of 5, so it ends in 0 or 5.
- $2q^4$ ends in 0 (if q^4 ends in 0 or 5) or in 2 (if q^4 ends in 1 or 6).
- p^4 ends in 0, 1, 5 or 6.

Adding a final digit from $\{0,1,5,6\}$ to one from $\{0,2\}$ gives one of $0,1,5,6,2,3,7,8$. The only ways to get 0 or 5 are $0+0$ and $5+0$. So $2q^4$ ends in 0, which means q^4 ends in 0 or 5, i.e. $5 \mid q$; and p^4 ends in 0 or 5, i.e. $5 \mid p$. This shows that p and q are divisible by 5.

(Or otherwise: by Fermat's little theorem $n^4 \equiv 1 \pmod 5$ when $5 \nmid n$, so $p^4 + 2q^4$ is congruent to $0+0$, $1+0$, $0+2$ or $1+2$ modulo 5, and only the first is 0.)

Write $p = 5p_1$ and $q = 5q_1$. Then

$$625(p_1^4 + 2q_1^4) = 5r^4 \quad \Longrightarrow \quad r^4 = 125(p_1^4 + 2q_1^4),$$

so $5 \mid r^4$, and since 5 is prime, $5 \mid r$. With $r = 5r_1$,

$$625r_1^4 = 125(p_1^4 + 2q_1^4) \quad \Longrightarrow \quad p_1^4 + 2q_1^4 = 5r_1^4.$$

So $(p/5,\ q/5,\ r/5)$ is again an integer solution of the same equation.

Exactly as in part (i), repeating this shows that 5^k divides each of p, q and r for every k, which is impossible for a non-zero integer. Hence the only integer solution is $p = q = r = 0$.

Checks

A computer search finds no solution other than $(0,0,0)$ of $a^3+3b^3=9c^3$ with $|a|,|b|,|c|\leq 60$, nor of $p^4+2q^4=5r^4$ with $|p|,|q|,|r|\leq 150$. Running through all 100 pairs of final digits for p and q confirms that p^4+2q^4 ends in 0 or 5 only when both final digits are 0 or 5. A quick hand check: $p=q=1$ gives $p^4+2q^4=3$, not a multiple of 5, as the table predicts.

Common slips

- Stopping after showing 3 (or 5) divides each unknown. That alone doesn't give zero; the point is that dividing out gives a solution of the *same* equation, so the argument repeats.
- Going from "$3 \mid a^3$" to "$3 \mid a$" without a reason. It works because 3 is prime; it would fail for, say, $9 \mid a^3$ implying $9 \mid a$ (try $a=3$).
- Forgetting negative integers when listing final digits. Fourth powers are even powers, so $p^4=|p|^4$ and the table covers every case.

Beyond STEP

Infinite descent goes back to Fermat, who used it to prove that $x^4+y^4=z^2$ has no solution in positive integers (and so the $n=4$ case of his Last Theorem). The idea used here, that an equation with no non-trivial solutions modulo a power of a prime can have no non-trivial integer solutions, grows into the study of p-adic numbers and the Hasse principle. Davenport's *The Higher Arithmetic* is a beautiful introduction to congruences and descent.

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